Prediction of Real-Gas PVT Behaviour, Thermodynamic Departure Functions, and Pure-Component Fugacity Using the Peng–Robinson Equation of State

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1. Introduction, Objectives, and Physical Scope

Accurate prediction of phase equilibria, volumetric behaviour, and thermodynamic properties of real fluids is a cornerstone of chemical process design. Ideal-gas models fail drastically at elevated pressures and near the vapour–liquid coexistence envelope. The objective of this project is to develop a scientifically rigorous, fully audited C++17 software application implementing the canonical Peng–Robinson (PR-1976) cubic equation of state for pure fluids. The software is engineered to predict: (1) Forward and inverse pressure–volume–temperature (PVT) behaviour; (2) Analytical compressibility factor roots with complete mechanical stability classification; (3) Residual enthalpy (H^{res}) and residual entropy (S^{res}); (4) Pure-component fugacity coefficients ($\ln \phi$, ϕ) and fugacities (f); (5) Reference-state total enthalpy and entropy referenced to the ideal gas at 298.15 K and 1.0 bar; and (6) Thermodynamic phase preference and saturation coexistence via rigorous equality of chemical potentials.

Physical Scope: The software strictly treats pure chemical components. Curated datasets are provided for eight industrial fluids: Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), n-Butane (C_4H_{10}), Nitrogen (N_2), Oxygen (O_2), Carbon Dioxide (CO_2), and Water Vapor (H_2O). All calculations are executed internally in strict SI units (Pa, K, m^3/mol, J/mol, J/(mol·K)).

2. Governing Mathematical Model and Thermodynamic Equations

2.1 The Peng–Robinson Cubic Equation of State (PR-1976)

The canonical pressure-explicit formulation proposed by Peng and Robinson (1976) expresses pressure P as the sum of a repulsive hard-sphere term and an attractive intermolecular dispersion term:

$$P(T, v) = [RT / (v - b)] - [a(T) / (v^2 + 2bv - b^2)]$$

where R = 8.314462618 J/(mol·K) is the universal gas constant (CODATA 2018), v is molar volume [m^3/mol], b is the co-volume parameter representing finite molecular volume, and $a(T)$ is the temperature-dependent attractive parameter. Enforcing the critical point inflection conditions $(\partial P/\partial v)_{T_c} = 0$ and $(\partial^2 P/\partial v^2)_{T_c} = 0$ yields the universal critical parameters:

$$a_c = \Omega_a^c (R^2 T_c^2 / P_c), \quad b = \Omega_b (R T_c / P_c), \quad \text{where } \Omega_a \approx 0.4572355289, \ \Omega_b \approx 0.0777960739$$

2.2 Temperature Dependence and Analytical Derivative

The attractive parameter accounts for temperature effects through the dimensionless scaling function $\alpha(T_r, \omega)$:

$$a(T) = a_c \alpha(T), \quad \alpha(T) = [\,1 + \kappa (1 - (T / T_c)^{1/2})\,]^2, \quad \kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2$$

where $T_r = T / T_c$ is the reduced temperature and ω is the Pitzer acentric factor. The exact analytical temperature derivative (da/dT), vital for departure functions, is given without empirical approximation by:

$$da/dT = -[\,a_c \kappa \alpha^{1/2}(T)\,] / [\,(T T_c)^{1/2}\,] \quad [Pa \cdot m^6/(mol^2 \cdot K)]$$

2.3 Dimensionless Compressibility Polynomial

Defining the compressibility factor $Z \equiv Pv / (RT)$ and the dimensionless parameters $A \equiv aP / (RT)^2$ and $B \equiv bP / (RT)$, the Peng–Robinson EOS transforms into a monic cubic polynomial:

$$Z^3 - (1 - B) Z^2 + (A - 3B^2 - 2B) Z - (AB - B^2 - B^3) = 0$$

At physical states, only roots satisfying $Z > B$ correspond to positive molar volumes exceeding the co-volume ($v > b$). Roots violating this boundary are strictly discarded as mathematical artefacts.

2.4 Residual Thermodynamic Departure Functions

Residual departure properties describe the difference between the actual thermodynamic state of a fluid and an ideal gas at the same temperature and pressure: $M^{res}(T, P) \equiv M(T, P) - M^{ig}(T, P)$. Using the fundamental Maxwell relations and integrating along an isothermal path from infinite volume to molar volume v:

$$H^{res}(T, P, Z) = RT\,(Z - 1) + [\,(T\,da/dT - a) / (2\sqrt{2}\,b)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,] \quad [J/mol]$$

$$S^{res}(T, P, Z) = R\,\ln(Z - B) + [\,(da/dT) / (2\sqrt{2}\,b)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,] \quad [J/(mol\cdot K)]$$

These closed-form equations (Peng & Robinson 1976, Eqs. 17–18) satisfy the rigorous thermodynamic identity $G^{res} = H^{res} - T\,S^{res} = RT \ln \phi$ identically.

2.5 Pure-Component Fugacity Coefficient and Fugacity

The fugacity coefficient $\phi \equiv f / P$ represents the dimensionless activity of the fluid relative to an ideal gas. From the fundamental thermodynamic relation $\ln \phi = \int_0^P [(Z - 1) / P]\,dP$, the exact PR expression is:

$$\ln \phi = (Z - 1) - \ln(Z - B) - [\,A / (2\sqrt{2}\,B)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,], \quad f = \phi \cdot P \quad [Pa]$$

In the ideal gas limit ($P \to 0$), $Z \to 1$, $B \to 0$, $\ln \phi \to 0$, and $\phi \to 1.0$ identically.

2.6 NIST Shomate Heat Capacity and Reference-State Total H and S

To calculate absolute state enthalpy $H(T, P)$ and entropy $S(T, P)$, the program integrates the authentic 5-parameter NIST Shomate isobaric heat capacity correlation for the ideal gas state ($t = T / 1000$):

$$C_p^{\circ}(t) = A + Bt + Ct^2 + Dt^3 + E / t^2 \quad [J/(mol\cdot K)]$$

Exact analytical integration yields the temperature departures ΔH^{ig} and ΔS^{ig}:

$$\Delta H^{ig}(T_{ref} \to T) = 1000\,[\,At + Bt^2/2 + Ct^3/3 + Dt^4/4 - E/t\,]_{t_{ref}}^{t} \quad [J/mol]$$

$$\Delta S^{ig}(T_{ref} \to T) = [\,A \ln t + Bt + Ct^2/2 + Dt^3/3 - E / (2t^2)\,]_{t_{ref}}^{t} \quad [J/(mol\cdot K)]$$

Reference State Convention: Total enthalpy and entropy are defined relative to the ideal gas reference state at T_{ref} = 298.15 K (25.0 °C) and P_{ref} = 100,000 Pa (1.0 bar), where $H_{total}(T_{ref}, P_{ref}) \equiv 0.0$ J/mol and $S_{total}(T_{ref}, P_{ref}) \equiv 0.0$ J/(mol·K). The total properties at arbitrary (T, P, Z) are:

$$H(T, P, Z) = \Delta H^{ig}(T_{ref} \to T) + H^{res}(T, P, Z) - H^{res}(T_{ref}, P_{ref}, Z_{ref})$$

$$S(T, P, Z) = \Delta S^{ig}(T_{ref} \to T) - R\,\ln(P / P_{ref}) + S^{res}(T, P, Z) - S^{res}(T_{ref}, P_{ref}, Z_{ref})$$

2.7 Low-Pressure Taylor Series Regularization

At near-vacuum pressures ($P < 10$ Pa), $B \to 0$, causing the quotient $\ln[(Z + (1+\sqrt{2})B)/(Z + (1-\sqrt{2})B)] / (2\sqrt{2}\,B)$ to approach 0 / 0. To prevent catastrophic floating-point cancellation, the software switches to a 4-term analytical Taylor series when $B/Z < 10^{-5}$:

$$I_L = (1 / Z) \cdot [\,1 - (B/Z) + (5/3)(B/Z)^2 - 3(B/Z)^3 + (33/5)(B/Z)^4\,]$$

This maintains double-precision continuity to 10^{-16} down to $P = 10^{-5}$ Pa.

2.8 Analytical Cubic Root Solver and Stability Classification

The cubic equation in Z is solved analytically using Cardano's formulation transformed into depressed form $y^3 + py + q = 0$ via the Tschirnhaus substitution $Z = y - c_2/3$. The discriminant $\Delta = q^2/4 + p^3/27$ governs the root structure: (1) If $\Delta > 0$, one real root exists (Cardano's real radical formula); (2) If $\Delta < 0$, three distinct real roots exist (Vieta's trigonometric formulation); and (3) If $\Delta = 0$, multiple real roots coalesce.

Mechanical Stability Criterion: For every physical root ($Z > B$), the isothermal pressure–volume derivative $(\partial P/\partial v)_T$ is evaluated analytically: $(\partial P/\partial v)_T = -RT/(v - b)^2 + a(T)(2v + 2b)/(v^2 + 2bv - b^2)^2$. States with $(\partial P/\partial v)_T < 0$ are **STABLE** (compression increases pressure). States with $(\partial P/\partial v)_T \geq 0$ are **UNSTABLE** (spinodal states; physically unobservable).

Phase Preference Criterion: In the three-root subcritical region, the smallest root (Z_{min}) is liquid-like, the largest root (Z_{max}) is vapour-like, and the intermediate root is mechanically unstable. Phase stability is decided by the minimum Gibbs free energy, which corresponds to the lower fugacity coefficient ϕ: $\Delta G^{res} = G_{liq}^{res} - G_{vap}^{res} = RT \ln(\phi_{liq} / \phi_{vap})$. If $\phi_{vap} < \phi_{liq}$ ($\Delta G^{res} > 0$), the vapour phase is preferred. If $\phi_{liq} < \phi_{vap}$ ($\Delta G^{res} < 0$), the liquid phase is preferred. If $\phi_{liq} \approx \phi_{vap}$ within 10^{-5} relative tolerance, saturated two-phase equilibrium exists.

3. Software Architecture and Implementation

The software is designed as a modular, header/source C++17 library (thermo_core) decoupled from user-interface logic. It relies exclusively on the C++ standard library (<cmath>, <vector>, <string>, <optional>, <iostream>) with zero external dependencies.

Module / Component	Source Files	Architectural Responsibility & Function
Constants	include/thermo/Constants.hpp	CODATA universal gas constant R, standard reference state constants, and analytical PR-1976 parameters.
Component & Database	src/Component.cpp src/ComponentDatabase.cpp	Validates physical parameters; stores curated database for 8 industrial gases; evaluates analytical Shomate integrals.
Cubic Solver	src/CubicSolver.cpp	Analytical Cardano/Vieta polynomial solver; filters duplicate roots; guarantees residuals ≤ 10^{-12}.
PR EOS Engine	src/PengRobinsonEOS.cpp	Forward/inverse PVT, $(\partial P/\partial v)_T$, departure functions H^{res}, S^{res}, fugacity ϕ, and phase preference.
Interactive CLI	src/cli/CLIController.cpp src/cli_main.cpp	Stream-decoupled interactive menu (`thermo_cli`), unit conversions (K/°C, Pa/bar/MPa), benchmark demo runner.
Automated Test Suite	tests/test_*.cpp (5 targets)	373 automated assertions covering cubic roots, Shomate quadrature, PVT reciprocity, Maxwell derivatives, and CLI.

3.1 Build System and Execution Instructions

The project is built using CMake 3.20+ with strict compiler warnings (`-Wall -Wextra -Wpedantic`):

cmake -B build -S . -DCMAKE_BUILD_TYPE=Release && cmake --build build

To run the full automated CTest suite (all 5 targets, 373 assertions):

ctest --test-dir build --output-on-failure

To run the interactive CLI or automated demonstration benchmarks:

./build/thermo_cli # Interactive mode
./build/thermo_cli --demo all # Automated demonstration benchmarks

4. Verification Results and Audited Numerical Benchmarks

All thermodynamic and volumetric calculations were independently verified against literature benchmarks and internal mathematical consistency checks. The table below reports values confirmed by direct execution of `thermo_cli`:

Case / State	Fluid & Phase	Z (PR-1976)	v [m^3/mol]	ϕ (Fugacity Coeff)	H^{res} [J/mol]	S^{res} [J/(mol·K)]	Status / Preference
Demo 1 300 K, 5.0 MPa	Methane (Gas)	0.901844	4.4990×10^{-4}	0.901385 (f=4.507 MPa)	–902.34	–2.1446	Single Phase (Stable)
Demo 2 300 K, 7.0 MPa	CO_2 (Dense Fluid)	0.198057	7.0575×10^{-5}	0.638285 (f=4.468 MPa)	–9374.37	–27.5150	Single Phase (Stable Liquid-like)
Demo 3 (Root 1) 320 K, 1.0 MPa	Propane (Liquid-like)	0.036453	9.6988×10^{-5}	1.239280 (f=1.239 MPa)	–14,854.34	–48.2035	Liquid Branch (Metastable)
Demo 3 (Root 2) 320 K, 1.0 MPa	Propane (Intermediate)	0.091826	2.4432×10^{-4}	1.385615 (f=1.386 MPa)	–8,210.60	–28.3698	**UNSTABLE** ($\partial P/\partial v$ > 0, Spinodal)
Demo 3 (Root 3) 320 K, 1.0 MPa	Propane (Vapour-like)	0.850555	2.2630×10^{-3}	0.868006 (f=0.868 MPa)	–1,131.04	–2.3575	**VAPOUR PREFERRED** (ΔG^{res} = +947.42 J/mol)

4.1 In-Depth Analysis of Multi-Root Propane Behaviour (Demo 3)

At 320 K and 1.0 MPa, Propane lies in the subcritical two-phase region ($\Delta = -1.083\times10^{-5} < 0$), generating three real roots: Root 1 (liquid-like, Z = 0.036453), Root 2 (intermediate, Z = 0.091826), and Root 3 (vapour-like, Z = 0.850555). The intermediate root exhibits $(\partial P/\partial v)_T = +1.8825\times10^{10}$ Pa/(m^3/mol) > 0, confirming it is mechanically unstable and located inside the spinodal zone. For the two mechanically stable roots, $\phi_{vap} = 0.868006 < \phi_{liq} = 1.239280$, producing $\Delta G^{res} = RT\ln(\phi_{liq}/\phi_{vap}) = +947.42$ J/mol > 0. The vapour phase possesses strictly lower chemical potential ($\mu_{vap} < \mu_{liq}$) and is selected as the stable equilibrium state.

4.2 Automated Testing and Independent Differential Maxwell Cross-Checks

The repository contains an automated test suite of **373 assertions across 5 CTest executables (100% pass rate)**. Circularity in validation was strictly avoided by implementing true independent differential checks: (1) **Gibbs–Helmholtz Cross-Check:** H^{res} was independently verified against numerical central differences $-RT^2(\partial \ln\phi/\partial T)_P$ with δT = 0.1 mK, matching to a relative error of 3.8×10^{-10}; (2) **Pressure Fugacity Differential:** $(\partial \ln\phi/\partial P)_T$ was matched to $(Z-1)/P$ to within 1.4×10^{-9}; and (3) **Numerical Quadrature:** Analytical Shomate integrals for ΔH^{ig} and ΔS^{ig} were cross-checked against an independent 1000-subinterval composite Simpson's 1/3 quadrature to within 10^{-6}.

5. Assumptions, Limitations, and Conclusion

Model Scope and Limitations: (1) *Pure Fluids Only:* Multicomponent mixtures and mixing rules are out of scope. (2) *Steam Reference State Limitation:* NIST Shomate gas-phase parameters for Water Vapor are valid only in [500.0, 1700.0] K. Because T_{ref} = 298.15 K lies outside this interval, total H and S are reported as N/A with clear diagnostics; PVT, fugacity, and residual departure functions remain fully operational. (3) *No Unphysical Extrapolation:* Calculating outside documented Shomate intervals throws exceptions rather than generating erroneous values. (4) *Cubic Liquid Volume Accuracy:* Near the critical point, two-parameter cubic EOSs typically underestimate liquid densities by 5–10% without volume translation (Peneloux).

Conclusion: The software provides a mathematically rigorous, verified, and robust engineering tool for pure-fluid thermodynamics. It successfully models complex non-ideal fluid behaviour, multi-root phase preference, and departure functions while maintaining high numerical precision.

References

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